THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE · HARD LEVEL CHAPTER TEST SET 3

Chapter 1: **Exploring the Investigative World of Science** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A claim on a product says '9 out of 10 users are happy' from a survey of 10 users all from the company. A concern is: 1

(a) too many users **(b)** no numbers
(c) biased and small sample **(d)** no product

2. Assertion (A): A conclusion should not go beyond the data collected. Reason (R): Data from one sample cannot always be generalised. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

3. To find the thickness of a sheet of paper with a metre ruler, the best idea is to: 1

(a) weigh a sheet **(b)** guess
(c) measure one sheet **(d)** measure a pile of many sheets and divide

4. Which action best reflects scientific honesty? 1

(a) guessing missing values **(b)** reporting a failed trial
(c) copying data **(d)** removing a trial that spoils the pattern

5. Assertion (A): The scientific method has a fixed sequence that must never change. Reason (R): Scientists often revisit earlier steps after unexpected results. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

6. In a study of the effect of salt on boiling point of water, which result would reject the hypothesis 'salt raises the boiling point'? 1

(a) higher boiling point for solution
(b) boiling at the same temperature for water and salt solution
(c) bubbles in both
(d) steam in both

7. In a table, a column 'Temperature (degree C)' has entries 20, 30, 40. This shows: 1

(a) no unit **(b)** wrong unit
(c) unit in every cell **(d)** unit given in the heading

8. Assertion (A): Safety goggles are worn when heating chemicals. Reason (R): Hot liquids may splash and damage the eyes. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

9. **Assertion (A):** A graph of the collected data can show how two variables are related. 1
Reason (R): A table lists data in rows and columns.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** The variable that a scientist deliberately changes in an experiment is the dependent variable. 1
Reason (R): The quantity that is measured as the outcome is the dependent variable.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. State the independent and dependent variable: 'How does the length of a wire affect its resistance?' 2

12. Why cannot a hypothesis be proved absolutely true by one experiment? 2

13. A thermometer shows 98 degrees C for boiling water at sea level. Name the likely error and its correction. 2

14. What is wrong in concluding that a coin is 'lucky' because heads came up 3 times in a row? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Explain why 'Hot water freezes quicker than cold water' is a hypothesis and not a fact. How can it be tested? 3

16. Why is the use of a stopwatch better than counting 'one-two-three'? 3

17. State three ways in which technology helps us in making more precise measurements. 3

OR

Why must a measuring cylinder be kept on a flat surface and read at the meniscus? 3

18. An experiment needs a long time to give a result, such as plant growth. How can the data be collected usefully? 3

OR

Give three examples of how asking questions has led to everyday inventions. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** Ravi tested whether the mass of the bob affects the time period of a simple pendulum. He used bobs of 20 g, 40 g and 60 g on a thread of the same length, and timed 20 oscillations in each case with a stopwatch. For the 20 g bob, 20 oscillations took 40 s. 5
(i) Why was the thread kept the same length?
(ii) Find the time period for the 20 g bob.
(iii) Why did he time 20 oscillations instead of one?
(iv) Name the independent and dependent variables.

20. State five features of a good scientific report and explain why each matters. 5

OR

Analyse: 'A cough syrup was given to 6 people and 5 got better. It works.' Point out weaknesses and how to improve. 5

END OF QUESTION PAPER — Total: 40 Marks